

Strategic Product Placement Analysis

Project Documentation

Project Description

This project aims to investigate the relationship between product positioning, sales performance, and consumer behavior. Using Tableau, we analyze sales data to uncover insights into how different positioning strategies impact sales and consumer preferences. By visualizing the data, we aim to provide actionable recommendations to optimize product positioning strategies and drive revenue growth.

A retail company wants to understand the impact of product positioning on its sales and consumer behavior. They have collected data on sales figures, product placement, and consumer demographics. They seek insights into which product positioning strategies are most effective in driving sales and how they can tailor their marketing efforts accordingly. Through data visualization in Tableau, the company hopes to gain actionable insights to improve its positioning strategies and increase revenue.

In retail environments, where a product is physically or digitally positioned — at eye level on a shelf, in a featured banner, or buried lower in a catalog page — can influence how often it is noticed, considered, and ultimately purchased. This project treats placement as a controllable variable and tests its effect against measurable outcomes such as revenue, order volume, and customer reach. The analysis groups every transaction into one of three placement tiers — Low Visibility, Medium, and Prime Placement — so that performance can be compared on a like-for-like basis across product categories, customer genders, and age bands.

The end goal is not just a set of charts but a decision-support tool: a retail merchandising or marketing team should be able to open the dashboard, filter to the placement strategy or segment they care about, and walk away with a clear, data-backed recommendation on where to invest shelf space, banner real estate, or promotional budget for the greatest revenue impact.

Skills Required

- Tableau (Business Intelligence Software)

Team Members

Team Member	Role
Muskan Shaik	Team Lead
Shaik Saniya	Member
Sri Varshini Velaga	Member
Pujitha Leela Kanamarlapudi	Member
Afrin Shaik	Member

Project Statistics

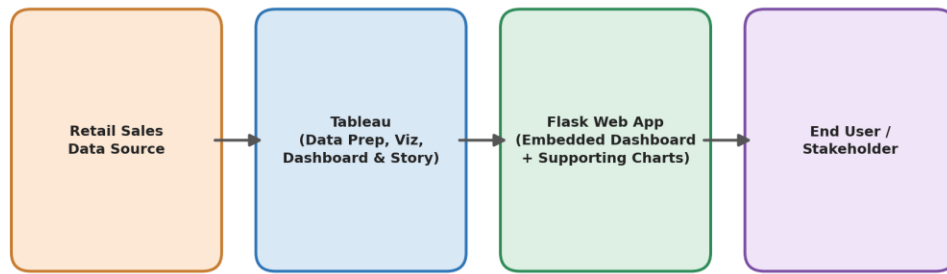
Metric	Value
Project Progress	100%
Total Epics	8
Total Tasks (Stories)	13
Total Subtasks	8
Complexity	Medium
Duration	5h 0m

Technical Architecture

The project follows an end-to-end analytics workflow: raw retail sales data is collected and extracted, cleaned and prepared, then modeled and visualized in Tableau as an interactive dashboard and an accompanying data story. Calculation fields and filters allow users to slice insights by product category, placement strategy (Low Visibility, Medium, and Prime Placement), gender, and age group. The finished Tableau dashboard and story are embedded into a lightweight Flask web application, giving stakeholders a single, browser-based interface to explore the analysis.

Architecturally, the solution is organized into four layers that map directly to the diagram below. The first layer is the data source itself — transaction-level retail sales records exported from the company's systems. The second layer is Tableau, which is responsible for every analytical step: connecting to the dataset, cleaning and shaping it, building the individual charts, assembling them into a responsive dashboard, and sequencing the key findings into a guided story. The third layer is the Flask web application, a thin presentation layer that does not perform any analysis itself; instead, it embeds the published Tableau dashboard using Tableau's JavaScript embedding API and arranges supporting KPI cards and charts around it for context. The fourth layer is the end user — a retail merchandising manager, marketer, or executive — who interacts with the dashboard entirely through the browser, with no need for Tableau Desktop or any specialized software installed locally.

This layered approach keeps the analytical logic centralized in Tableau (where it is easiest to maintain and extend) while the Flask layer keeps the access experience simple, shareable, and brand-consistent. Because the dashboard is embedded rather than recreated in code, any update made in Tableau — a new chart, an adjusted calculation, a refreshed dataset — automatically reflects in the web application without further development work.

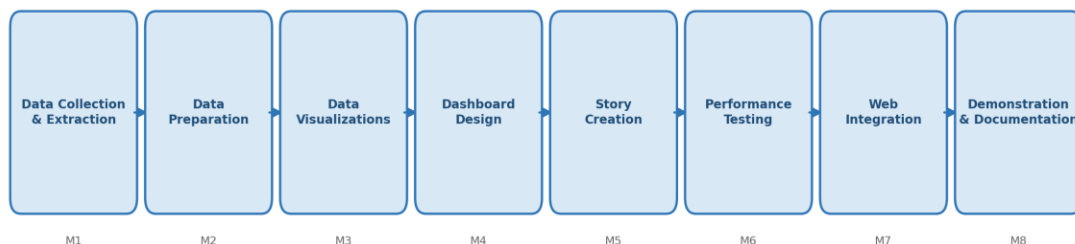


Pre-requisites

- Tableau Desktop / Tableau Public knowledge: data connections, calculated fields, and dashboard design.
- Basic understanding of retail sales datasets (transactions, product categories, customer demographics).
- Flask framework knowledge for embedding the dashboard into a web interface.
- HTML and CSS skills for the supporting web pages.
- Familiarity with data preparation and cleaning concepts.

Project Workflow

To accomplish this project, the following activities are completed across each milestone:



Milestone 1: Data Collection & Extraction from Database

- Collect the dataset
- Connect data with Tableau

Milestone 2: Data Preparation

- Prepare the data for visualization

Milestone 3: Data Visualizations

- Determine the number of unique visualizations required

Milestone 4: Dashboard

- Responsive design of the dashboard

Milestone 5: Story

- Define the number of scenes in the story

Milestone 6: Performance Testing

- Utilization of data filters
- Number of calculation fields
- Number of visualizations / graphs

Milestone 7: Web Integration

- Embed the dashboard and story with the UI using Flask

Milestone 8: Project Demonstration & Documentation

- Record an explanation video for the project end-to-end solution
- Prepare project documentation with step-by-step project development procedure

Milestone Details

Milestone 1: Data Collection & Extraction from Database

This milestone focuses on sourcing and connecting the retail sales dataset that underpins the entire analysis. Every downstream chart, calculation, and recommendation depends on the quality and completeness of the data gathered here, so this stage is treated as the foundation of the project rather than a quick first step.

Activity 1.1: Collect the Dataset

- Gather retail sales data covering transactions, product categories, placement strategy, and customer demographics (gender, age group).
- Ensure the dataset spans a sufficient time period (the current dataset covers 2023 to early 2024) to identify meaningful sales trends.
- Confirm that each transaction record carries a placement tag (Low Visibility, Medium, or Prime Placement) so that positioning impact can be measured directly rather than inferred.

Activity 1.2: Connect Data with Tableau

- Import the dataset into Tableau Desktop / Tableau Public.
- Verify field types (dates, numeric measures, categorical dimensions) are correctly recognized on connection.

- Spot-check row counts and a sample of records after import to confirm the connection has not dropped or duplicated data.

Milestone 2: Data Preparation

In this milestone, the raw data is cleaned and structured so it is ready for visualization. Clean, consistent data at this stage prevents misleading charts later — a single mislabeled category or missing value can distort an entire visualization, so this step is given particular care.

Activity 2.1: Prepare the Data for Visualization

- Clean inconsistent or missing values in the sales records.
- Create derived fields such as total order amount, average order value, and quantity per order.
- Group products into categories (Electronics, Clothing, Beauty) and tag each record with its placement strategy.
- Standardize demographic fields (gender, age group) into consistent buckets so they can be reliably grouped and filtered on the dashboard.

Milestone 3: Data Visualizations

This milestone covers building the individual charts that make up the analysis. Each visualization is chosen to answer a specific business question rather than purely for visual variety — for example, a line chart for trend-over-time questions, and a stacked bar for composition-and-comparison questions.

Activity 3.1: Determine the Number of Unique Visualizations

- Sales Trend Over Time line chart, comparing performance across placement strategies.
- Product Category vs Placement stacked bar chart.
- Monthly Sales Trend line/area chart for a single year view.
- Sales by Category donut chart.
- Sales by Gender bar chart.
- Sales by Age Group bar chart.
- Category Performance summary table (Revenue and Orders by category).

Together, these seven visualizations cover three analytical angles: positioning impact over time (Sales Trend, Category vs Placement), product-level performance (Category donut, Category Performance table), and customer-level performance (Gender, Age Group). This combination allows the dashboard to answer not just “which placement works best” but also “for which products” and “for which customers.”

Milestone 4: Dashboard

A consolidated, interactive Tableau dashboard is designed to bring all visualizations together into a single, coherent view rather than a set of disconnected charts.

Activity 4.1: Responsive Design of the Dashboard

- Arrange the Sales Trend Over Time and Product Category vs Placement charts together for quick comparison.
- Add filter controls for Low Visibility Placement, Medium Placement, and Prime Placement so users can isolate a strategy.
- Ensure the dashboard layout adapts cleanly when embedded inside the web application.
- Apply consistent color coding for placement strategies and product categories across every chart so the dashboard reads as a single, unified view rather than separate reports.

Milestone 5: Story

A Tableau Story is built to walk stakeholders through the key insights in a guided narrative, so that someone unfamiliar with the dashboard can still understand the headline findings in a few minutes without exploring every filter combination themselves.

Activity 5.1: Define the Number of Scenes in the Story

- Scene 1 — Overview: introduce total revenue, transactions, and top category.
- Scene 2 — Placement Comparison: highlight how Prime Placement drives a sharper sales trend than Medium and Low Visibility placements.
- Scene 3 — Category Insights: break down performance across Electronics, Clothing, and Beauty.
- Scene 4 — Customer Demographics: present sales split by gender and age group.
- Scene 5 — Recommendations: summarize actionable positioning recommendations.

Each scene reuses the same underlying dashboard views with a different filter state and annotation, so the story stays visually consistent with the interactive dashboard while telling a clear, linear narrative.

Milestone 6: Performance Testing

Before integration, the dashboard and story are tested for accuracy and responsiveness, so that every number a stakeholder sees can be trusted and every filter behaves predictably.

Activity 6.1: Utilization of Data Filters

- Test the Low Visibility / Medium / Prime Placement filter across every chart for consistent behavior.
- Confirm that combining the placement filter with category or demographic filters produces correct, non-conflicting results.

Activity 6.2: Number of Calculation Fields

- Validate calculated fields such as Average Order Value (Total Revenue / Transactions) and Average Quantity per Order.
- Cross-check calculated totals against a manual spot-check of the raw data to confirm formula correctness.

Activity 6.3: Number of Visualizations / Graphs

- Confirm each of the seven visualizations renders correctly with live data and updates when filters change.
- Check dashboard load and filter-response time so the experience stays smooth when embedded in the web application.

Milestone 7: Web Integration

The finished Tableau dashboard and story are embedded into a Flask-based web application so the analysis can be shared as a single, browsable solution without requiring stakeholders to log into Tableau directly.

Activity 7.1: Embed Dashboard and Story with UI Using Flask

- Build a Flask app with Dashboard, Story, and About navigation tabs.
- Embed the published Tableau Public dashboard inside the Dashboard page using Tableau's embed/JavaScript API.
- Surface top-level KPI cards (Total Revenue, Transactions, Avg Order Value, Unique Customers, Top Category, Avg Qty/Order) above the embedded view.
- Add supporting charts (Monthly Sales Trend, Sales by Category, Sales by Gender, Sales by Age Group, Category Performance table) below the embedded dashboard.
- Style the page with simple, brand-neutral HTML/CSS so the embedded Tableau content remains the visual focus.

Milestone 8: Project Demonstration & Documentation

The final milestone packages the project for handover and review, making sure the work is reproducible and understandable by someone outside the immediate project team.

Activity 8.1: Record an Explanation Video

- Record an end-to-end walkthrough demonstrating the dashboard, story, and web integration.

Activity 8.2: Prepare Project Documentation

- Document the step-by-step project development procedure, covering data collection through web integration.

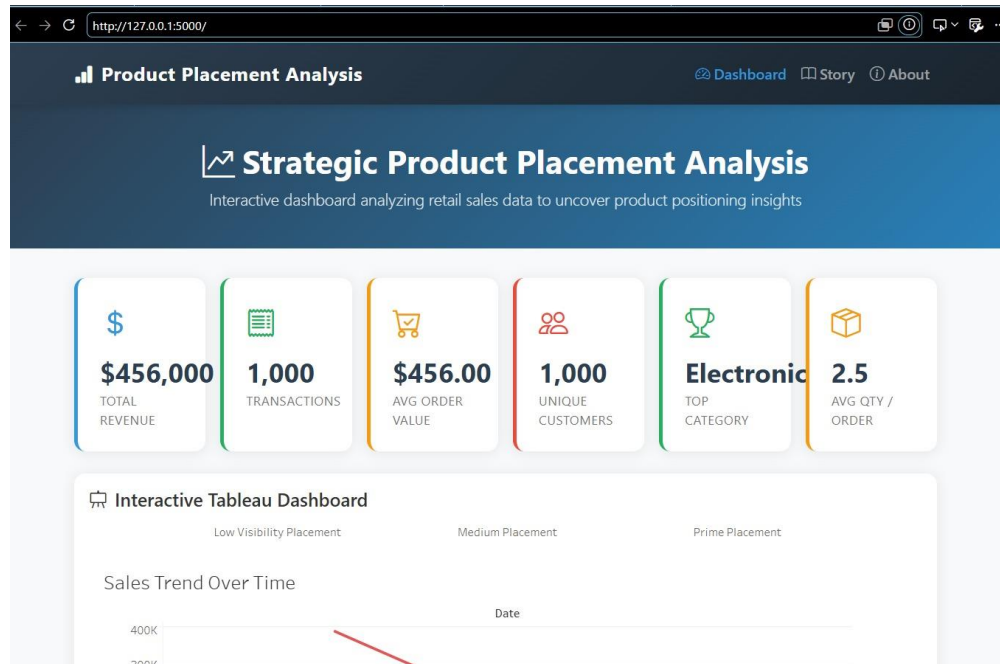
Exploring the Web Application

This section walks through the web application screen by screen, explaining what each part of the page shows and how it connects back to the underlying Tableau analysis.

Dashboard Page

The Dashboard page presents the KPI summary cards followed by the embedded Tableau dashboard, allowing users to filter by placement strategy and review the sales trend and category-versus-placement charts. The header introduces the project with a short title and subtitle, and the navigation bar at the top right lets users move between the Dashboard, Story, and About pages without leaving the application.

The six KPI cards immediately below the header give an at-a-glance summary of overall performance before the user drills into any chart: total revenue, transaction count, average order value, number of unique customers, the top-performing category, and the average quantity per order. These figures are computed across the full dataset, so they serve as a fixed reference point that the more detailed charts below can be compared against once filters are applied.



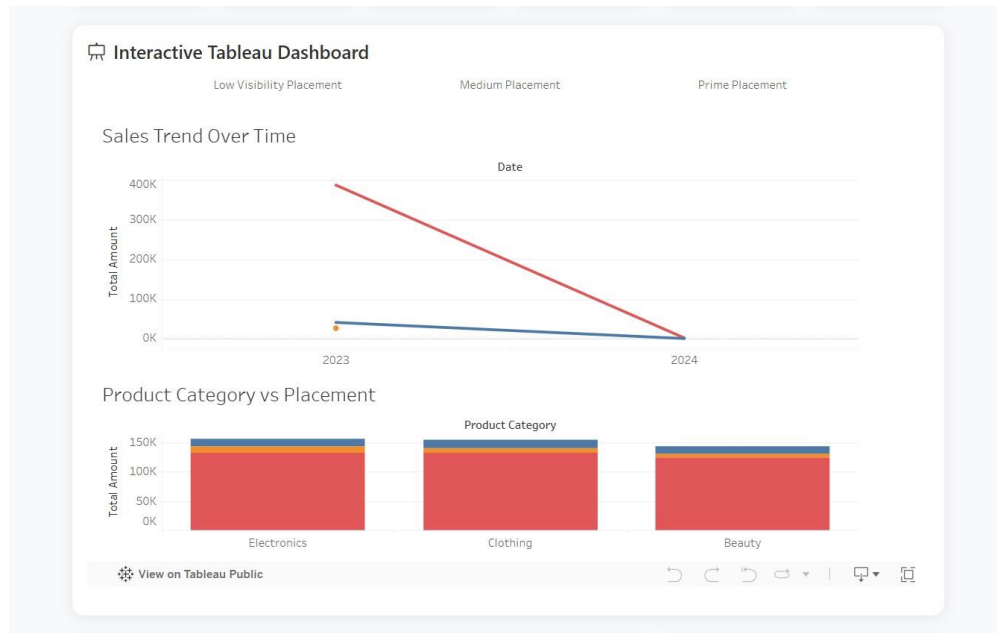
Metric	Value	Notes
Total Revenue	\$456,000	Across all transactions
Transactions	1,000	Total order count
Avg Order Value	\$456.00	Revenue / Transactions
Unique Customers	1,000	Distinct buyers
Top Category	Electronics	Highest revenue category
Avg Qty / Order	2.5	Items per order

Interactive Tableau Dashboard

The embedded dashboard shows the Sales Trend Over Time chart, comparing Low Visibility, Medium, and Prime Placement strategies, alongside a Product Category vs Placement breakdown for Electronics, Clothing, and Beauty. The three placement toggles above the Sales Trend chart let the user isolate a single strategy or compare them side by side, and the chart updates instantly because it is the live, published Tableau view rather than a static image.

Reading the Sales Trend Over Time chart, Prime Placement shows a markedly steeper trajectory in 2023 compared with Medium and Low Visibility Placement, which is the central finding the rest of the dashboard and the accompanying story are built to support. The Product

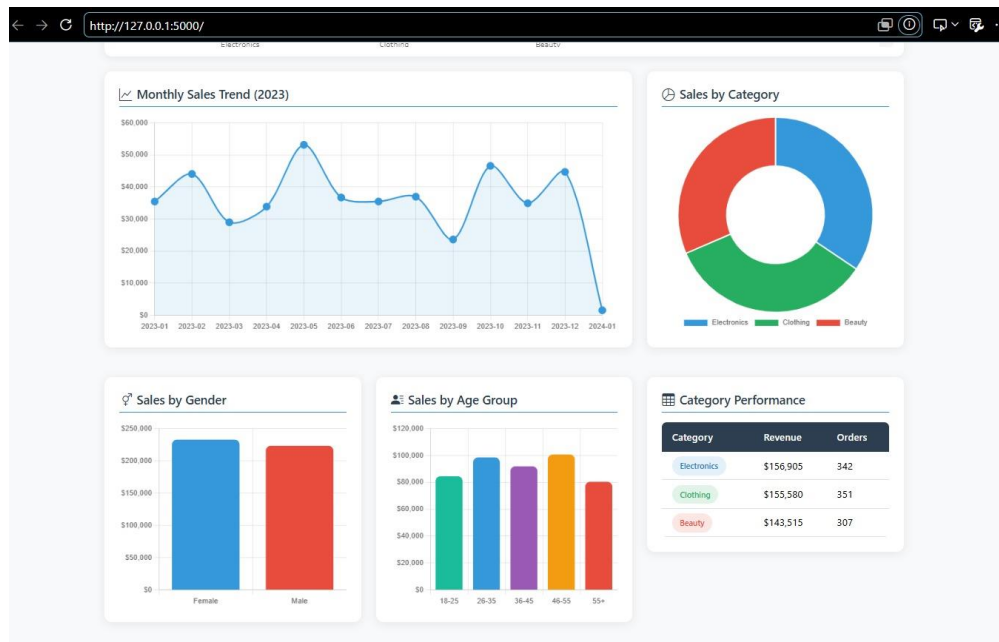
Category vs Placement chart beneath it breaks the same data down by product category, making it possible to check whether the placement effect holds consistently for Electronics, Clothing, and Beauty, or whether it is concentrated in one category.



Supporting Analytics Section

Below the main dashboard, the page surfaces additional supporting visuals: a Monthly Sales Trend for 2023, a Sales by Category donut chart, Sales by Gender and Sales by Age Group bar charts, and a Category Performance summary table. These five panels exist to answer follow-up questions that naturally arise after reviewing the main placement comparison — namely, how sales move month to month, which category contributes the most revenue, and which customer segments are driving demand.

The Monthly Sales Trend chart shows a seasonal, somewhat volatile pattern across 2023, with a peak around May and a smaller secondary peak in October and November, useful context when planning when to schedule Prime Placement campaigns. The Sales by Category donut chart shows a fairly even three-way split between Electronics, Clothing, and Beauty, while the Sales by Gender and Sales by Age Group charts show that demand is broadly balanced across genders and concentrated in the 26-35 and 46-55 age bands. The Category Performance table closes the section with the exact revenue and order counts behind the donut chart, giving stakeholders precise figures to cite alongside the visual summary.



Category	Revenue	Orders
Electronics	\$156,905	342
Clothing	\$155,580	351
Beauty	\$143,515	307

Conclusion

The Strategic Product Placement Analysis project delivers an interactive Tableau-powered solution that helps a retail company understand how product positioning strategies influence sales performance and consumer behavior. By collecting and preparing sales data, building targeted visualizations, and assembling them into a responsive dashboard and guided story, the project surfaces clear, actionable insights — such as the stronger sales trend associated with Prime Placement versus Medium and Low Visibility placements, and category- and demographic-level performance differences. Embedding the dashboard into a Flask web application makes these insights accessible through a simple, shareable interface.

Beyond the headline placement finding, the project demonstrates a complete, repeatable analytics workflow: from raw data extraction, through cleaning and calculated-field design, to chart construction, dashboard assembly, narrative storytelling, structured testing, and finally web delivery. This workflow is intentionally modular — each milestone produces an output that the next milestone consumes — so it can be revisited and extended without reworking the entire project from scratch.

Looking ahead, the solution can be extended with predictive analytics to forecast the sales impact of future placement decisions, deeper customer segmentation, and automated refresh of the underlying dataset. Additional opportunities include A/B-testing new placement tiers directly within the dashboard, incorporating profit margin alongside revenue to capture true business

impact, and adding role-based views so merchandising, marketing, and executive stakeholders each see the metrics most relevant to their decisions.